



Technische
Universität
Braunschweig

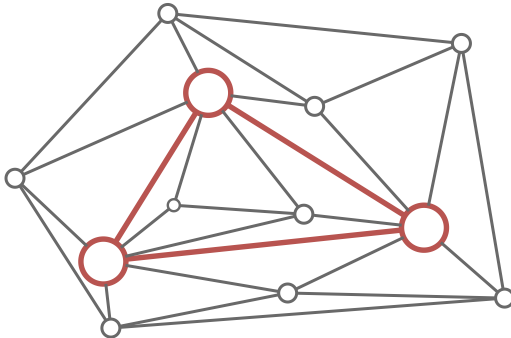


Algorithmische Methoden zu gradbeschränkten Triangulierungen

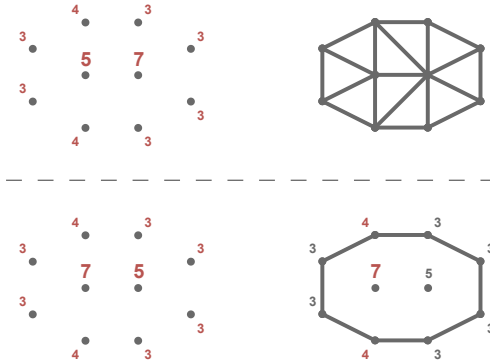
Fabian Alich, 3. September 2025

Praktische Relevanz

FA: keine Ahnung was ich sonst noch Praktisch sagen soll.



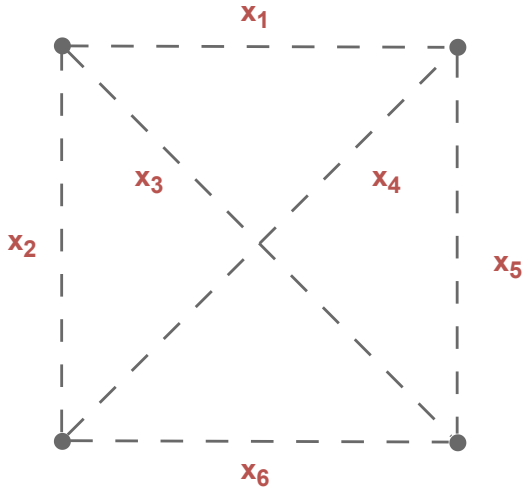
Das Problem



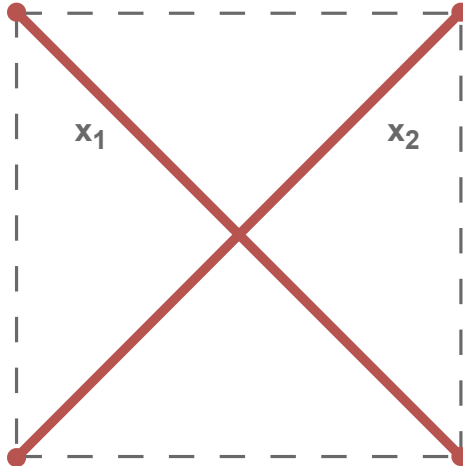
Teil I

Lösungsansätze

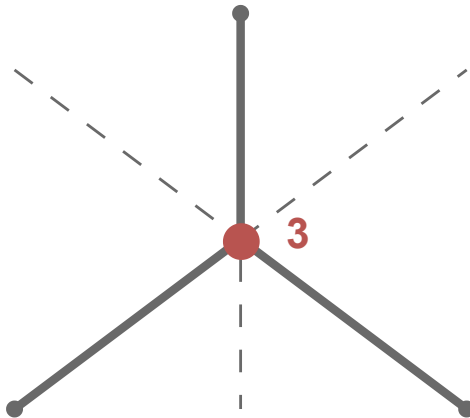
Kantenansatz



Überschneidungsbedingung



Gradbeschränkung



Maximale Kanten

FA: Formal raus?

$$\text{Anzahl Kanten} \cdot 2 \stackrel{!}{=} \sum \text{Sollgrad}$$

Gradbeschränkung

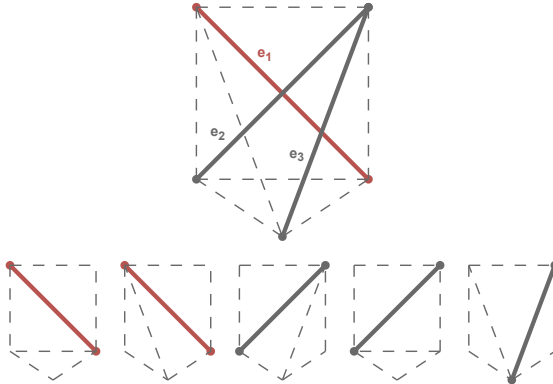
FA: Durchschnitt

Gradbeschränkung

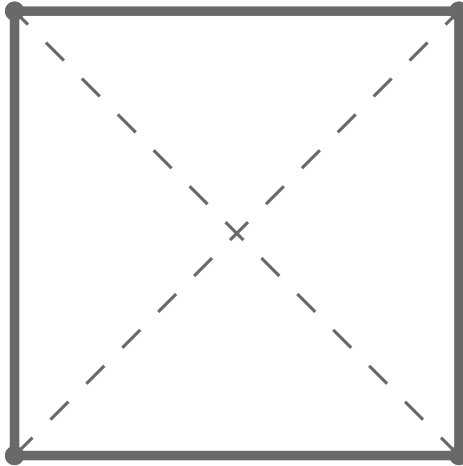
FA: Max min

alternative Kantenbedingung

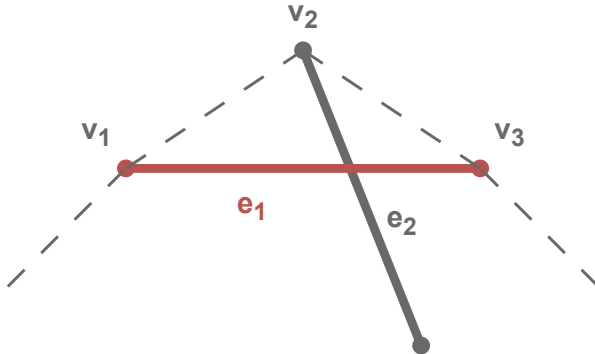
FA: sollte ich hier eine Referenz auf euer Paper machen?



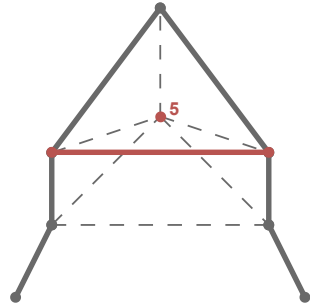
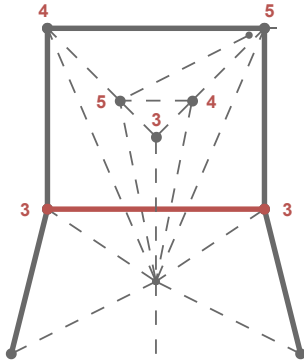
Hülle Fixieren



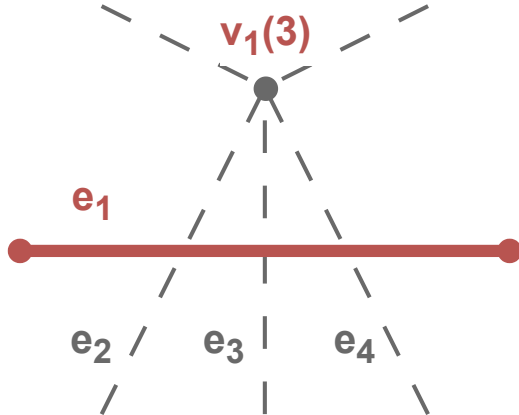
Kanten Fixieren / Ausschließen



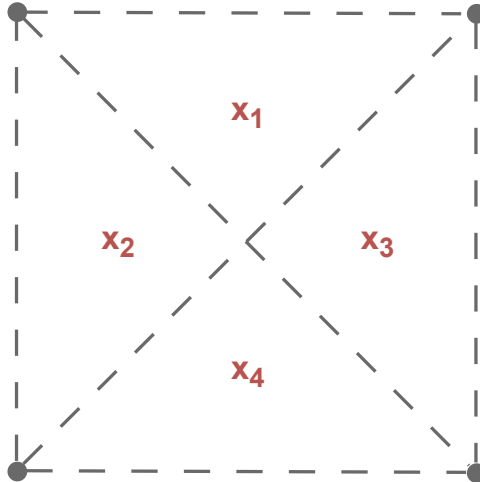
Kanten Ausschließen



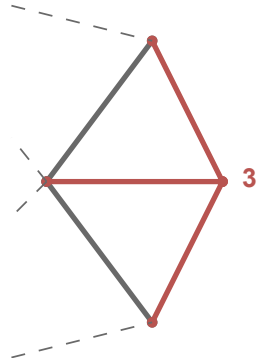
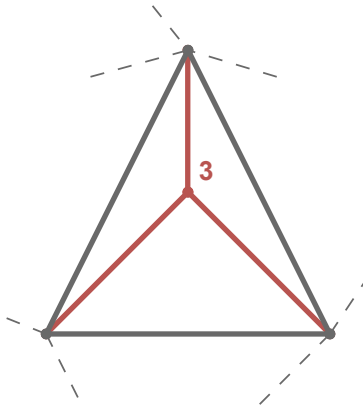
Kanten Ausschließen



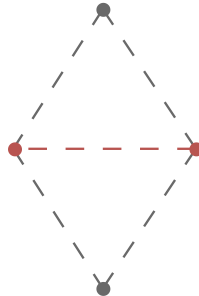
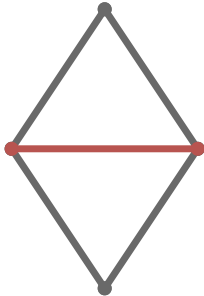
Dreiecksansatz



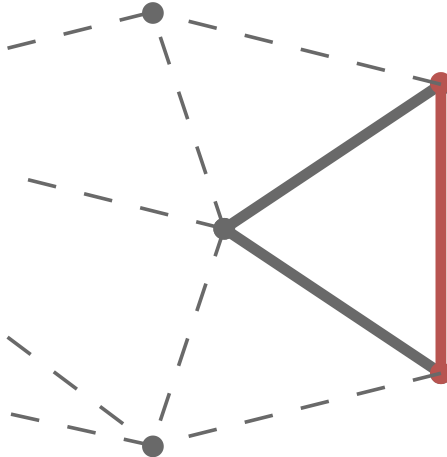
Gradbeschränkung



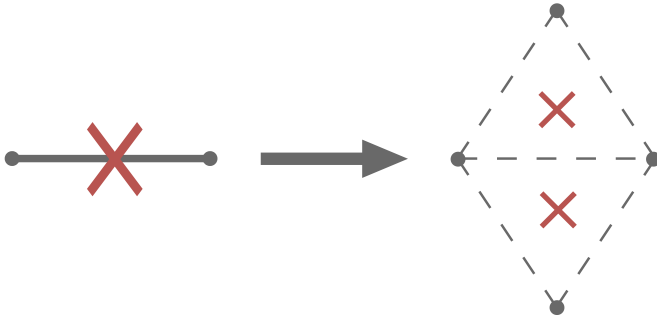
Überschneidungsbedingung



Überschneidungsbedingung



Dreiecks Ausschließen



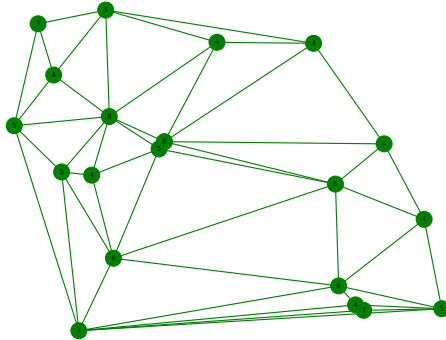
Subset

FA: ich habe jetzt subset in den Diagrammen drin, ist aber schlecht. Soll ich das dann hier erklären oder wie mache ich das am besten? Einfach nur auf die Arbeit verweisen?

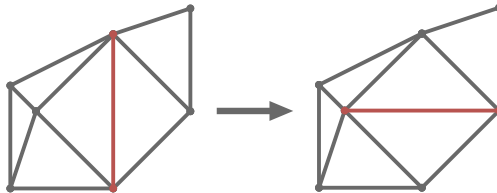
Teil II

Instanzgenerierung

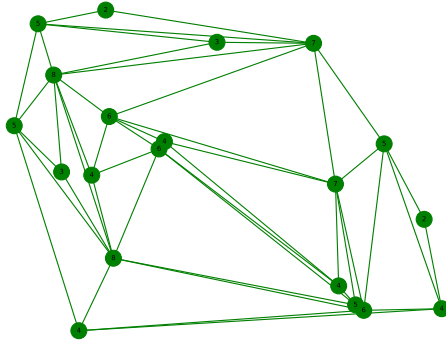
Delaunay



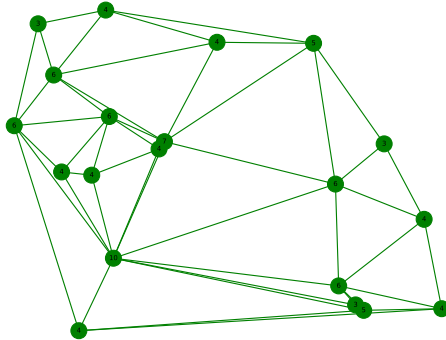
Flip



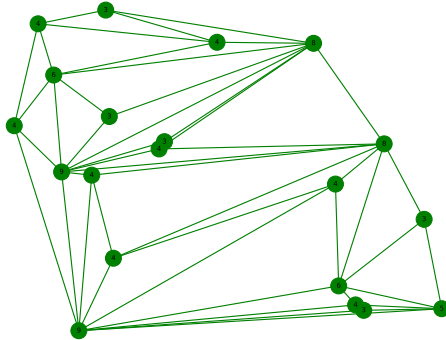
Delaunay-Flips



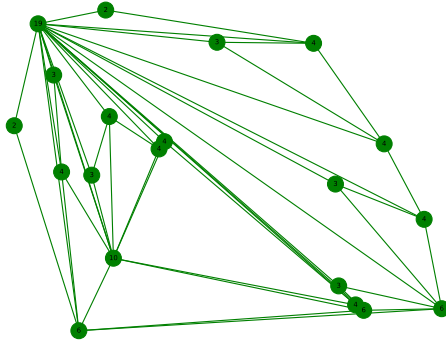
Greedy



Random



Iterativ



Move

FA: Bild erstellen

Change

FA: Bild erstellen

Teil III

Auswertung

Solver



Gurobi

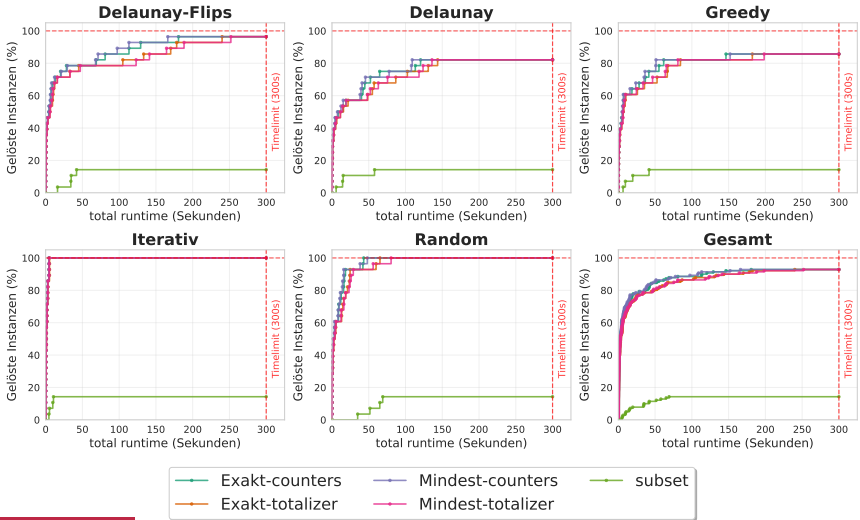


OR-Tools

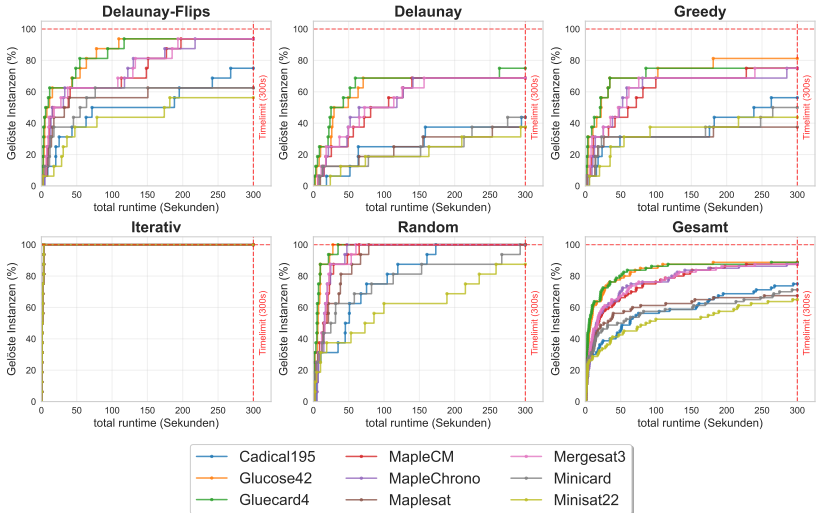


PySAT

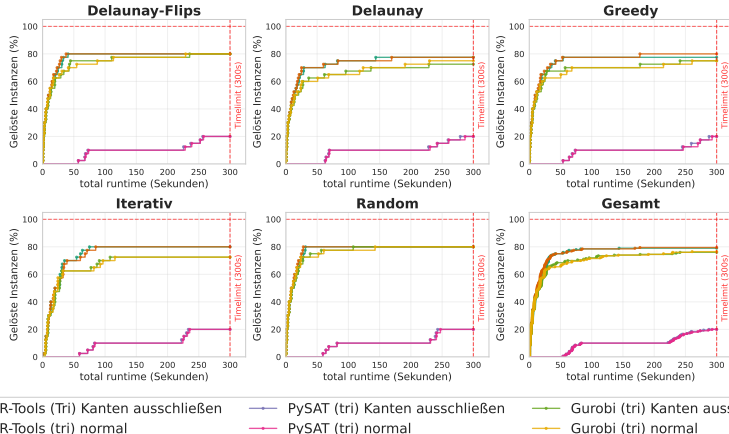
Grad Codierung



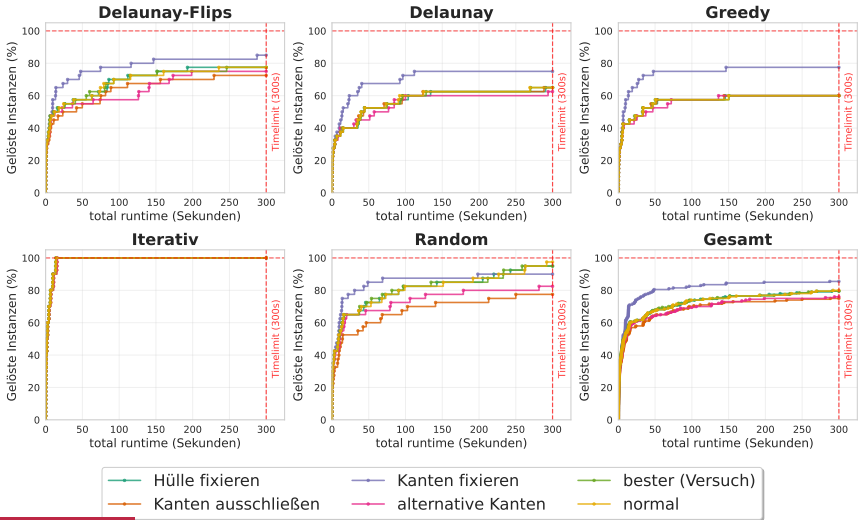
PySAT Solver



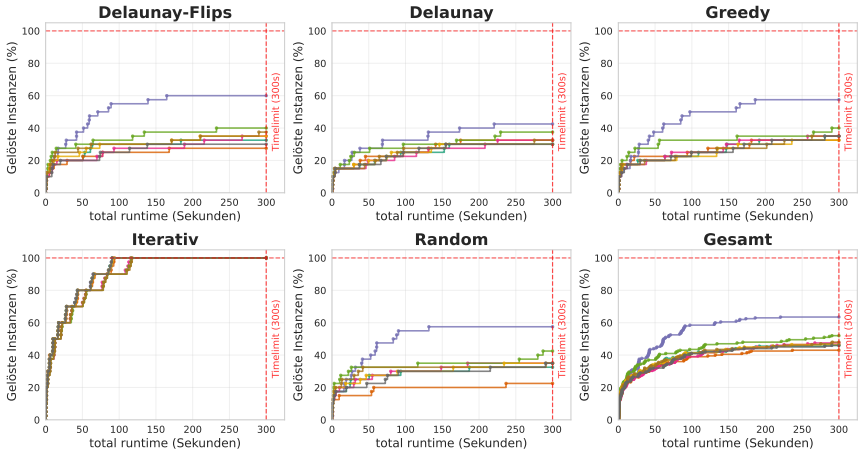
Dreiecke Bedingungen



PySAT Bedingungen

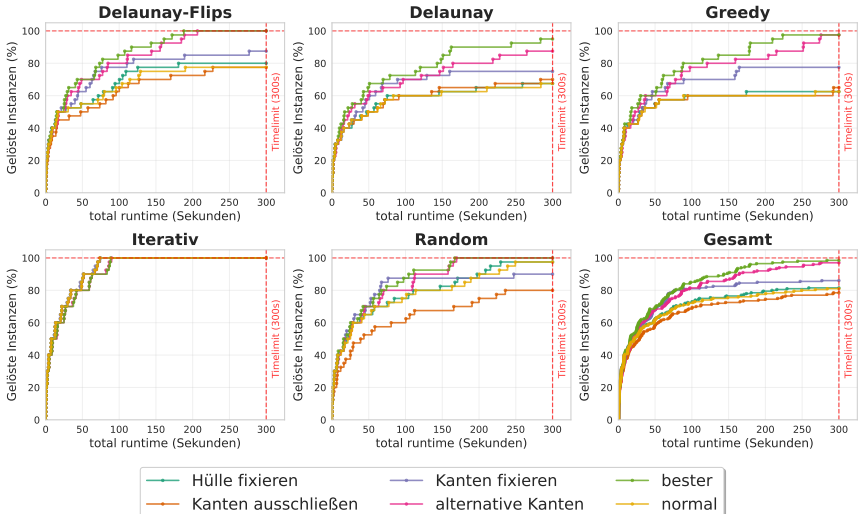


Gurobi Bedingungen

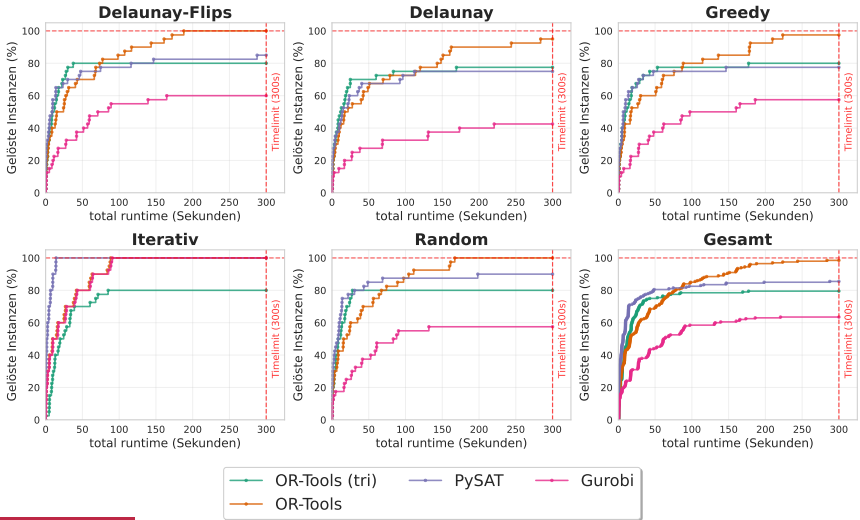


- Hülle fixieren
- Kanten ausschließen
- Kanten fixieren
- alternative Kanten
- bester (Versuch 1)
- bester (Versuch 2)
- bester (Versuch 3)
- normal

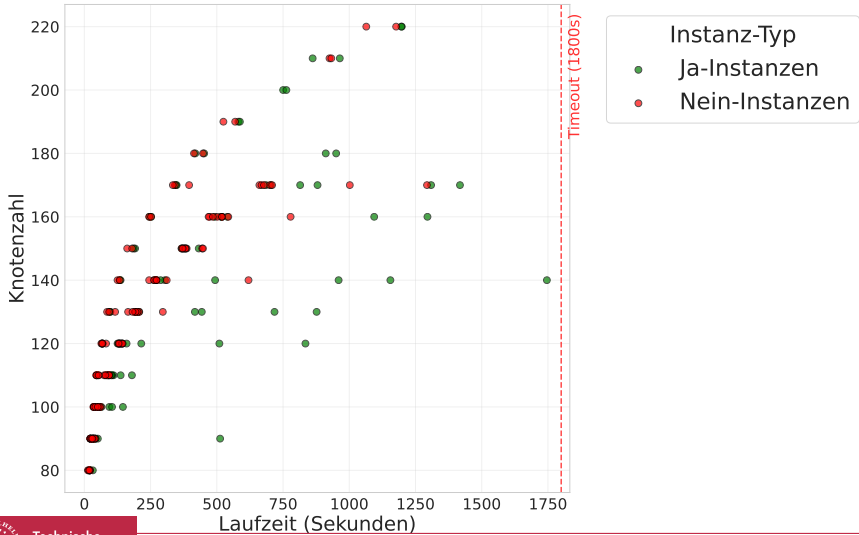
OR-Tools Bedingungen



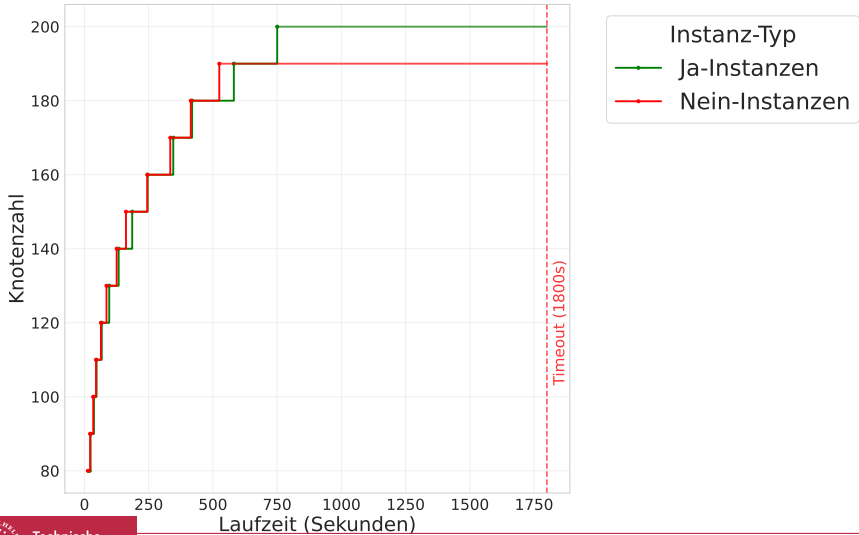
Solver Vergleich



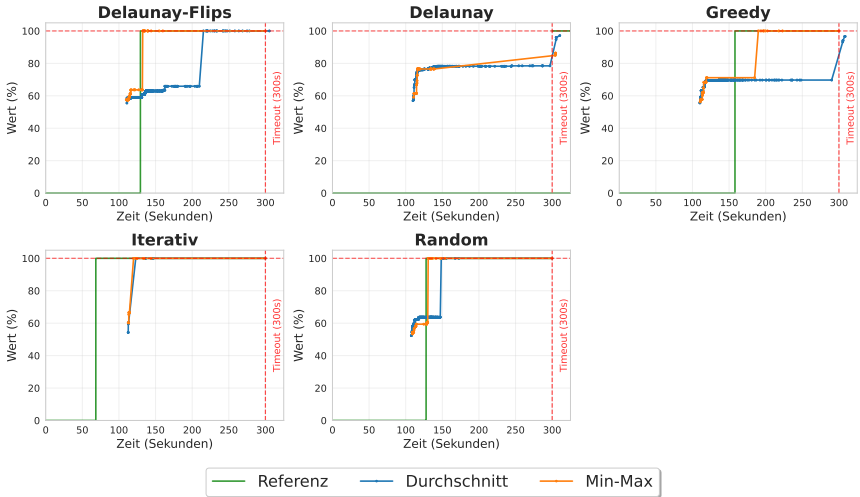
Größerer Timeout



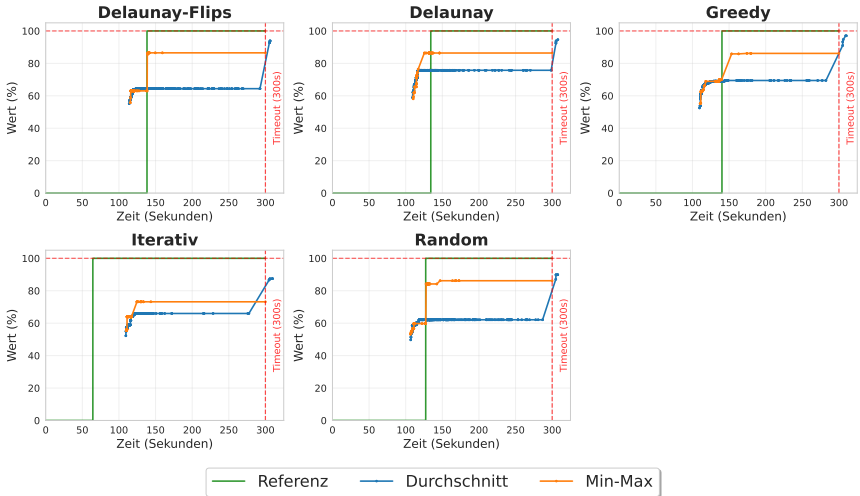
Ja Nein getrennt



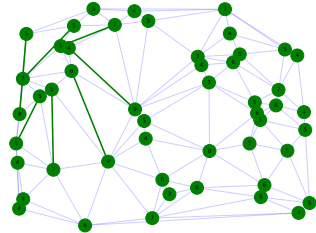
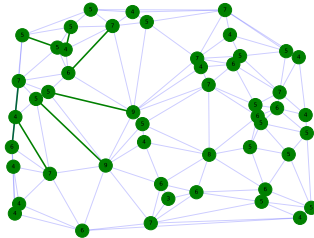
Ja Instanz



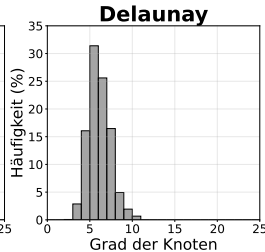
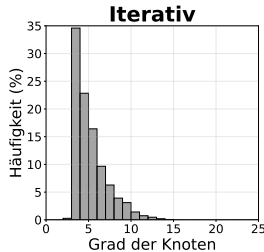
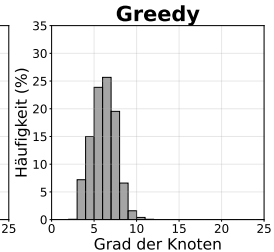
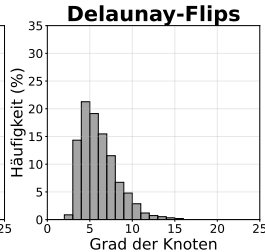
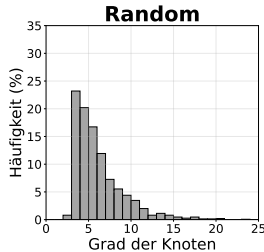
Nein Instanz



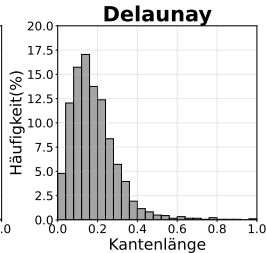
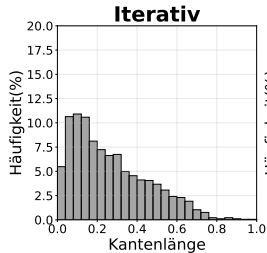
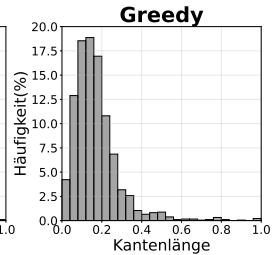
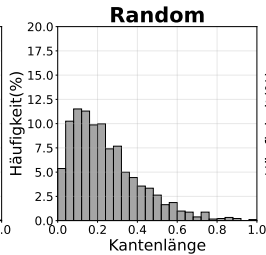
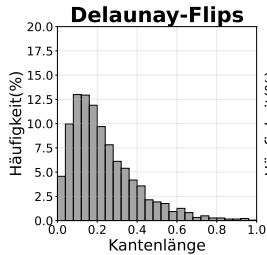
Anzahl Lösungen



Gradverteilung



Kantenlängen

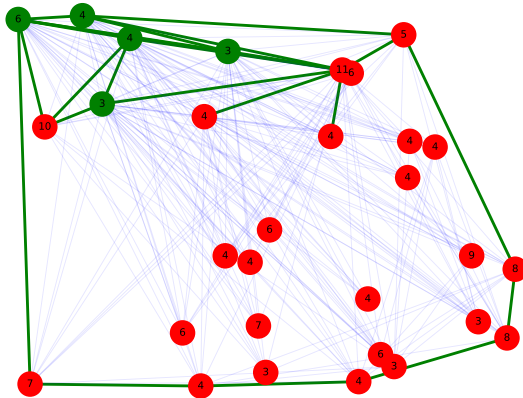


CaDiCaL Simplified Satisfiability Solver

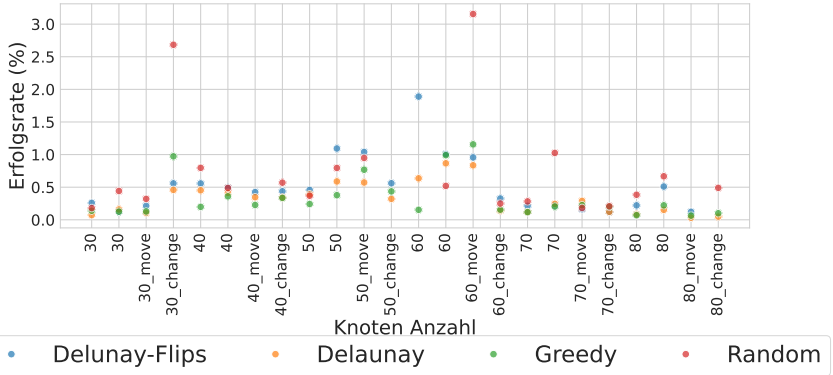
Teilinstanz

Isolierte Kante Teilinstanz einfügen

Teilinstanz



Propagation



Teil IV

Fazit und Ausblick

Fazit

OR-Tools Hülle fixieren Kanten Fixieren alternative Kantenbedingung

Fazit

Bild von Instanze mit 220 Knoten

Fazit

Bild von Optimierungsproblem mit 97 Prozent

Fazit

Bild wo eine Propagation möglich war

Ausblick

- Skalierbarkeit
- Arbeitsspeicherverbrauch
- Dreiecksansatz Schwierigkeit Instanzen
- Optimierungen Kombinationen
- Propagation erweitern
- normalverteilte Triangulierung
- Problem abwandeln
 - Kanten vorgeben
 - Polygone